Native PENK Amendment 5 — the extracellular environment changes the analysis

Scope: native (unmodified) human proenkephalin, P01210, 267 aa. Companion to Amendment 2, which covered granule processing only. **Basis:** the peptidase evidence established in engPENK Amendment 4, applied to the native panel.

§0 What changes, and why

Amendment 2 analysed native PENK as a **granule** problem: PC1/PC3 and PC2 cut at dibasic sites, CPE trims basic residues, PAM amidates a Gly-extended precursor. That is correct and stands. But it stops at the granule membrane. Once secreted, every product enters a compartment with a different pH, no Ca^{2+} gradient, and a different enzyme set — **NEP, ACE and APN on the cell surface**. Three consequences, none of which were in Amendment 2:

1. The panel is not 25 peaks. It is **49**, because ACE generates a second tier of products.
2. Two enkephalin peaks are **route-ambiguous** — the same mass arises granule-side and extracellular-side.
3. The MS method needs a stated resolving power, because the extended panel contains a 1.27 Da pair.

Amendment 2's own §0 retraction (core 5 is a PAM substrate; cassettes 4/7 are native architecture) is unaffected.

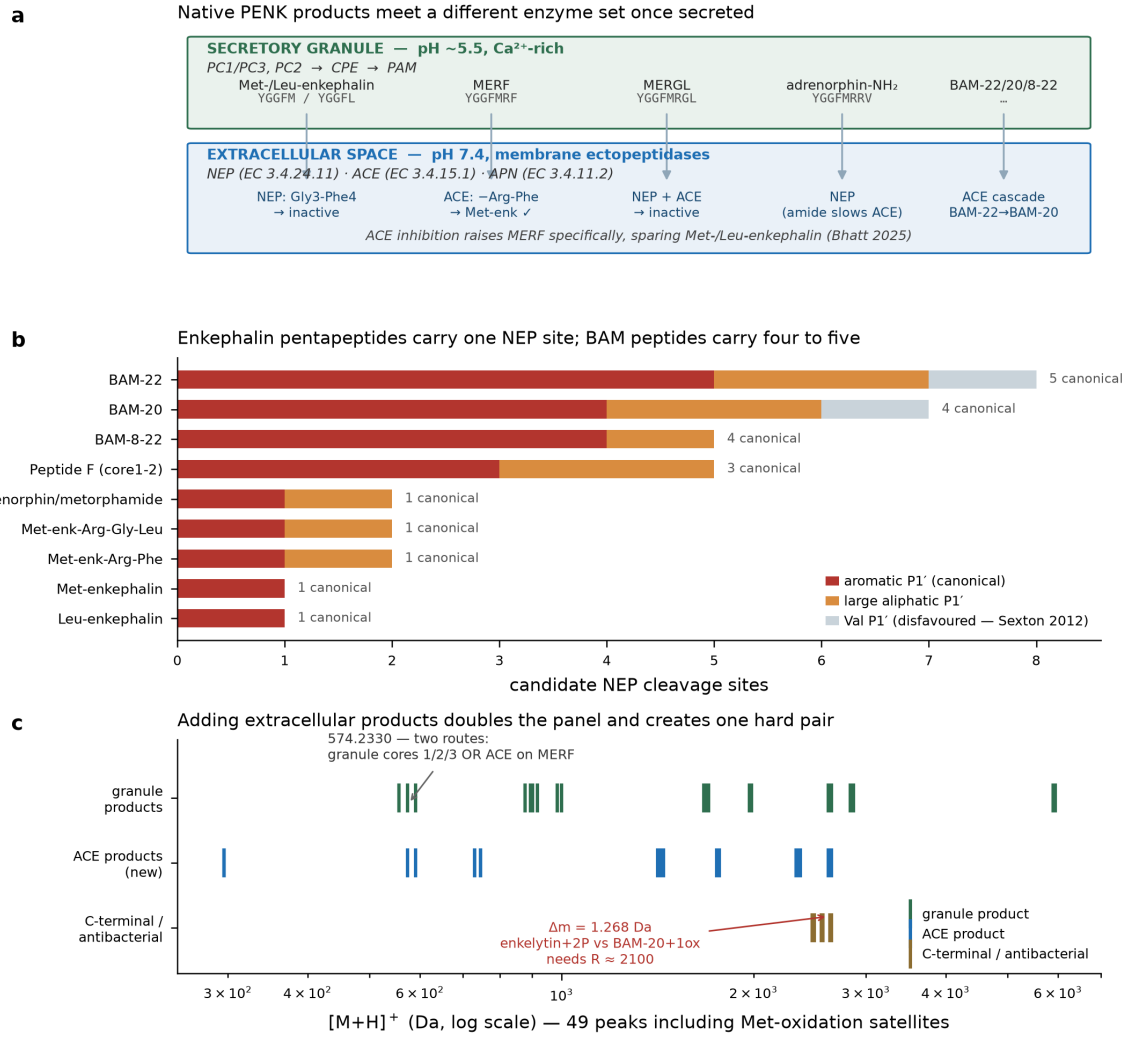


Figure. (a) Native PENK products cross from the granule enzyme set (PC1/PC2, CPE, PAM) into the extracellular set (NEP, ACE, APN); fates differ per species. (b) Tiered NEP subsite scoring — pentapeptides carry one canonical site, BAM peptides four to five. (c) The 49-peak extended panel, with the enkelytin+2P / BAM-20+1ox pair (Δm = 1.268 Da) and one dual-route mass annotated.

§1 The NEP site count, corrected

Amendment 2 inherited the flat rule — every bond followed by a hydrophobic residue counted as a neprilysin site. That rule is wrong (see engPENK Amendment 4 §0 R1): the S1′ pocket is shaped by Phe563 and prefers **aromatic** P1′ residues, and a valine at P1′ leaves an unfavourable gap on binding (Sexton et al. 2012). Re-scored in tiers:

species	canonical (aromatic P1′)	large aliphatic P1′	Val P1′ (disfavoured)
Leu-enkephalin `YGGFL`	**1**	0	0
Met-enkephalin `YGGFM`	**1**	0	0
MERF `YGGFMRF`	**1**	1	0
MERGL `YGGFMRGL`	**1**	1	0
adrenorphin `YGGFMRRV`-NH_2	**1**	1	0
Peptide F `YGGFMKRYGGFMKK`	**3**	2	0
BAM-8-22	**4**	1	0
BAM-20	**4**	2	1
BAM-22	**5**	2	1

The pentapeptides carry exactly one canonical site — `Gly3-Phe4`, the classical enkephalinase bond. The **BAM peptides carry four to five**, which is the quantitative reason they are short-lived extracellularly and why granule content and secreted content are not the same population.

§2 ACE doubles the panel

ACE is a dipeptidyl carboxypeptidase; it removes a C-terminal dipeptide, and it acts on amidated substrates too (Bernstein et al. 2013). Applied to every native species:

substrate	ACE removes	product	product $[M+H]^+$	identity
`YGGFMRF` (MERF)	Arg-Phe	`YGGFM`	574.2330	**Met-enkephalin — activating**
`YGGFMRGL` (MERGL)	Gly-Leu	`YGGFMR`	730.3341	novel fragment
`YGGFMRRV`-NH_2	Arg-Val-NH_2	`YGGFMR`	730.3341	novel fragment
`YGGFM`	Phe-Met	`YGG`	296.1241	inactive
`YGGFL`	Phe-Leu	`YGG`	296.1241	inactive
BAM-22	Tyr-Gly	BAM-20	2618.2656	**already in the panel**
BAM-8-22	Tyr-Gly	`VGRPEWWMDYQKR`	1750.8482	novel fragment
Peptide F	Lys-Lys	`YGGFMKRYGGFM`	1413.6442	novel fragment

The MERF→Met-enkephalin conversion is ACE-dependent in vivo, not a test-tube artefact: ACE inhibition raises extracellular MERF specifically while leaving Met- and Leu-enkephalin unaffected (Bhatt et al. 2025).

Two route ambiguities follow directly. A peak at 574.2330 is Met-enkephalin whether it came from granule cores 1/2/3 or from ACE acting on secreted MERF. A peak at 2618.2656 is BAM-20 whether it was a granule intermediate or ACE product of secreted BAM-22. Neither can be resolved by mass. **Resolving them requires a captopril (± thiorphan) arm**, not a better mass spectrometer.

§3 The C-terminal region Amendment 2 omitted

Amendment 2's panel covered the opioid cores and BAM intermediates. It did not include the precursor's C-terminal acidic region, which is a major secreted product:

species	residues	$[M+H]^+$
enkelytin region, free acid	PENK(237-258) `FAEALPSDEEGESYSKEVPEME`	2473.0599

species	residues	[M+H]⁺
enkelytin, +1 phospho	"	2553.0262
enkelytin, +2 phospho	"	2632.9925
Peptide B	PENK(210-258)	5893.7624

This region carries three Ser and one Tyr, so the phospho-forms are the species to expect, not the free acid. **This is where the panel gets hard**: enkelytin+2P at 2632.9925 sits **1.268 Da** from BAM-20+1ox at 2634.2605 — a resolving power of **≈2100** at z=1, and the same requirement at z=2 and z=3. Any instrument below that will merge the two. Every other pair in the 49-peak panel is either resolved or a same-sequence duplicate.

§4 What this means for the construct comparison

The engineered cassette is judged against native background, so the background matters:

- **Native Met-enkephalin has one canonical NEP site. So does every cassette product.** The earlier claim that the cassette was less stable was retracted in engPENK Amendment 4; this analysis confirms the native side of that comparison independently.
- **Both are ACE substrates and both give an activating conversion.** ACE turns native MERF into Met-enkephalin; it turns the cassette 7 analogue `YGGFEVVGRF` into the target octapeptide `YGGFEVVG` (827.3934). The same chemistry, on both sides.
- **BAM-8-22 remains the shared peak** (1970.9330, engPENK Amendment 3 §0) — it lies downstream of the swapped core and is produced identically by native and construct. Its ACE product `VGRPEWWMDYQKR` at 1750.8482 is shared for the same reason. Add it to the shared-peak list, not the marker list.

§5 Bench actions

1. **Extend the MS target list to 49 peaks** — see `native_penk_extended_ms_panel.csv`. The 24 new entries are ACE products and the C-terminal/phospho forms.
2. **Specify R ≥ 3000** on the method (1.4× margin over the 2100 requirement) or exclude the 2620–2640 window from quantitation.
3. **Add a captopril arm** to any experiment quantifying Met-enkephalin or BAM-20. Without it, granule-derived and ACE-derived peaks are not separable.
4. **Add thiorphan** if BAM peptide recovery is low — four to five canonical NEP sites predicts rapid loss, and low recovery is expected rather than a handling failure.
5. **Look for the phospho-enkelytin forms**, not the free acid.

§6 Checklist

- [ ] MS target list replaced with the 49-peak version
- [ ] Instrument resolving power confirmed ≥3000, or 2620–2640 excluded
- [ ] Captopril (ACE) arm added where Met-enkephalin or BAM-20 is quantified
- [ ] Thiorphan (NEP) arm available for BAM recovery troubleshooting
- [ ] BAM-8-22 **and** its ACE product moved to the shared-peak list
- [ ] Amendment 2's flat NEP counts superseded by §1

References

1. Sexton T. et al. (2012) PLoS ONE 7:e32343 — NEP S1′ pocket, Phe563, aromatic P1′ preference. doi: 10.1371/journal.pone.0032343
2. Bernstein K.E. et al. (2013) Pharmacol Rev 65:1 — ACE as dipeptidyl carboxypeptidase; amidated substrates. doi:10.1124/pr.112.006809
3. Bhatt et al. (2025) Commun Biol — ACE inhibition raises MERF selectively. doi:10.1038/s42003-025-08790-6
4. Ozawa A. et al. (2007) Anal Biochem — in-vitro PC2/CPE/PAM cascade.